

## Lecture 2: Singular value decomposition (SVD)



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## 1. Singular value decomposition

- **Definition:** Let  $m$  and  $n$  be arbitrary positive integers ( $m \geq n$  or  $m < n$ ). Given  $\mathbf{A} \in \mathbb{C}^{m \times n}$ , not necessarily of full rank, a *singular value decomposition (SVD)* of  $\mathbf{A}$  is a factorization

$$\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^*,$$

where  $\mathbf{U} \in \mathbb{C}^{m \times m}$  is unitary,  $\mathbf{V} \in \mathbb{C}^{n \times n}$  is unitary, and  $\mathbf{\Sigma} \in \mathbb{R}^{m \times n}$  is diagonal. In addition, it is assumed that the diagonal entries  $\sigma_i$  of  $\mathbf{\Sigma}$  are nonnegative and in nonincreasing order; that is

$$\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_p \geq 0,$$

where  $p = \min\{m, n\}$ . We have  $\mathbf{\Sigma} = \text{diag}\{\sigma_1, \sigma_2, \dots, \sigma_p\} \in \mathbb{R}^{m \times n}$ .

### Theorem 1 (Existence of SVD)

*Every matrix  $\mathbf{A} \in \mathbb{C}^{m \times n}$  has a singular value decomposition.*

**Proof.** Assume  $\mathbf{A} \neq \mathbf{0}$ ; otherwise we can take  $\mathbf{\Sigma} = \mathbf{0}$  and let  $\mathbf{U}$  and  $\mathbf{V}$  be arbitrary unitary matrices. Next, we use induction on  $m$  and  $n$  to prove the existence of SVD for the case  $m \geq n$  (consider  $\mathbf{A}^*$  if  $m < n$ ): Assume that an SVD exists for any  $(m-1) \times (n-1)$  matrix and prove it for any  $m \times n$  matrix.

(i) The basic step:  $m \geq n = 1$ .

Write  $\mathbf{A} = \mathbf{u}_1 \mathbf{\Sigma}_1 \mathbf{V}^*$  with  $\mathbf{u}_1 = \mathbf{A} / \|\mathbf{A}\|_2$ ,  $\mathbf{\Sigma}_1 = \|\mathbf{A}\|_2$  and  $\mathbf{V} = 1$ . Choose  $\mathbf{U}_c$  such that  $\mathbf{U} = [\mathbf{u}_1 \quad \mathbf{U}_c] \in \mathbb{C}^{m \times m}$  is unitary. Let  $\mathbf{\Sigma} = [\mathbf{\Sigma}_1 \quad \mathbf{0}]^T \in \mathbb{R}^{m \times 1}$ . Then  $\mathbf{A}$  has an SVD  $\mathbf{A} = \mathbf{U} \mathbf{\Sigma} \mathbf{V}^*$ .

(ii) The induction step:  $m \geq n > 1$ .

Let  $\tilde{\mathbf{v}}_1 \in \mathbb{C}^n$  be a unit (i.e.,  $\|\tilde{\mathbf{v}}_1\|_2 = 1$ ) eigenvector corresponding to the eigenvalue  $\lambda_{\max}(\mathbf{A}^* \mathbf{A})$ . Then we have  $\|\mathbf{A} \tilde{\mathbf{v}}_1\|_2 = \|\mathbf{A}\|_2 > 0$ . Let  $\tilde{\mathbf{u}}_1 = \mathbf{A} \tilde{\mathbf{v}}_1 / \|\mathbf{A} \tilde{\mathbf{v}}_1\|_2$ , which is a unit vector. Choose  $\tilde{\mathbf{U}}_c$  and  $\tilde{\mathbf{V}}_c$  such that  $\tilde{\mathbf{U}} = [\tilde{\mathbf{u}}_1 \quad \tilde{\mathbf{U}}_c] \in \mathbb{C}^{m \times m}$  and  $\tilde{\mathbf{V}} = [\tilde{\mathbf{v}}_1 \quad \tilde{\mathbf{V}}_c] \in \mathbb{C}^{n \times n}$  are unitary.

Now we have

$$\tilde{\mathbf{U}}^* \mathbf{A} \tilde{\mathbf{V}} = \begin{bmatrix} \tilde{\mathbf{u}}_1^* \\ \tilde{\mathbf{U}}_c^* \end{bmatrix} \mathbf{A} \begin{bmatrix} \tilde{\mathbf{v}}_1 & \tilde{\mathbf{V}}_c \end{bmatrix} = \begin{bmatrix} \tilde{\mathbf{u}}_1^* \mathbf{A} \tilde{\mathbf{v}}_1 & \tilde{\mathbf{u}}_1^* \mathbf{A} \tilde{\mathbf{V}}_c \\ \tilde{\mathbf{U}}_c^* \mathbf{A} \tilde{\mathbf{v}}_1 & \tilde{\mathbf{U}}_c^* \mathbf{A} \tilde{\mathbf{V}}_c \end{bmatrix}.$$

We note that

$$\tilde{\mathbf{u}}_1^* \mathbf{A} \tilde{\mathbf{v}}_1 = \frac{(\mathbf{A} \tilde{\mathbf{v}}_1)^* (\mathbf{A} \tilde{\mathbf{v}}_1)}{\|\mathbf{A} \tilde{\mathbf{v}}_1\|_2} = \|\mathbf{A} \tilde{\mathbf{v}}_1\|_2 = \|\mathbf{A}\|_2,$$

$$\tilde{\mathbf{U}}_c^* \mathbf{A} \tilde{\mathbf{v}}_1 = \tilde{\mathbf{U}}_c^* \tilde{\mathbf{u}}_1 \|\mathbf{A} \tilde{\mathbf{v}}_1\|_2 = \mathbf{0},$$

and

$$\tilde{\mathbf{u}}_1^* \mathbf{A} \tilde{\mathbf{V}}_c = \frac{(\mathbf{A} \tilde{\mathbf{v}}_1)^*}{\|\mathbf{A} \tilde{\mathbf{v}}_1\|_2} \mathbf{A} \tilde{\mathbf{V}}_c = \frac{\tilde{\mathbf{v}}_1^* \mathbf{A}^* \mathbf{A} \tilde{\mathbf{V}}_c}{\|\mathbf{A} \tilde{\mathbf{v}}_1\|_2} = \frac{\lambda_{\max}(\mathbf{A}^* \mathbf{A}) \tilde{\mathbf{v}}_1^* \tilde{\mathbf{V}}_c}{\|\mathbf{A} \tilde{\mathbf{v}}_1\|_2} = \mathbf{0}.$$

Let

$$\sigma_1 := \|\mathbf{A}\|_2.$$

Then we have

$$\tilde{\mathbf{U}}^* \mathbf{A} \tilde{\mathbf{V}} = \begin{bmatrix} \sigma_1 & \mathbf{0} \\ \mathbf{0} & \tilde{\mathbf{U}}_c^* \mathbf{A} \tilde{\mathbf{V}}_c \end{bmatrix}.$$

By the induction hypothesis, we know that the  $(m-1) \times (n-1)$  matrix  $\tilde{\mathbf{U}}_c^* \mathbf{A} \tilde{\mathbf{V}}_c$  has an SVD:

$$\tilde{\mathbf{U}}_c^* \mathbf{A} \tilde{\mathbf{V}}_c = \hat{\mathbf{U}} \hat{\Sigma} \hat{\mathbf{V}}^*.$$

It follows from  $\sigma_1 = \|\mathbf{A}\|_2$ , unitary invariance of  $\|\cdot\|_2$ , and

$$\tilde{\mathbf{U}}^* \mathbf{A} \tilde{\mathbf{V}} = \begin{bmatrix} \sigma_1 & \mathbf{0} \\ \mathbf{0} & \hat{\mathbf{U}} \hat{\Sigma} \hat{\mathbf{V}}^* \end{bmatrix} = \begin{bmatrix} 1 & \mathbf{0} \\ \mathbf{0} & \hat{\mathbf{U}} \end{bmatrix} \begin{bmatrix} \sigma_1 & \mathbf{0} \\ \mathbf{0} & \hat{\Sigma} \end{bmatrix} \begin{bmatrix} 1 & \mathbf{0} \\ \mathbf{0} & \hat{\mathbf{V}} \end{bmatrix}^*$$

that  $\sigma_1 \geq \|\hat{\Sigma}\|_2$ . Now it is straightforward to show that

$$\mathbf{A} = \tilde{\mathbf{U}} \begin{bmatrix} 1 & \mathbf{0} \\ \mathbf{0} & \hat{\mathbf{U}} \end{bmatrix} \begin{bmatrix} \sigma_1 & \mathbf{0} \\ \mathbf{0} & \hat{\Sigma} \end{bmatrix} \begin{bmatrix} 1 & \mathbf{0} \\ \mathbf{0} & \hat{\mathbf{V}} \end{bmatrix}^* \tilde{\mathbf{V}}^* =: \mathbf{U} \begin{bmatrix} \sigma_1 & \mathbf{0} \\ \mathbf{0} & \hat{\Sigma} \end{bmatrix} \mathbf{V}^*$$

is an SVD of  $\mathbf{A}$ . □

- Let  $\mathbf{U} = [\mathbf{u}_1 \quad \mathbf{u}_2 \quad \cdots \quad \mathbf{u}_m]$  and  $\mathbf{V} = [\mathbf{v}_1 \quad \mathbf{v}_2 \quad \cdots \quad \mathbf{v}_n]$ .  
 $\{\sigma_i^2, \mathbf{u}_i\}$  and  $\{\sigma_i^2, \mathbf{v}_i\}$  are eigenpairs of  $\mathbf{A}\mathbf{A}^*$  and  $\mathbf{A}^*\mathbf{A}$ , respectively:

$$\mathbf{A}\mathbf{A}^*\mathbf{u}_i = \sigma_i^2\mathbf{u}_i, \quad \mathbf{A}^*\mathbf{A}\mathbf{v}_i = \sigma_i^2\mathbf{v}_i, \quad i = 1, 2, \dots, p.$$

$\{\sigma_i\}$  are called the *singular values* of  $\mathbf{A}$ ,  $\{\mathbf{u}_i\}$  are called the *left singular vectors*, and  $\{\mathbf{v}_i\}$  are called the *right singular vectors*:

$$\mathbf{u}_i^*\mathbf{A} = \sigma_i\mathbf{v}_i^*, \quad \mathbf{A}\mathbf{v}_i = \sigma_i\mathbf{u}_i, \quad i = 1, 2, \dots, p.$$

## Theorem 2

*The set of singular values  $\{\sigma_i\}$  is uniquely determined and invariant under unitary matrix multiplication.*

## Theorem 3

*If  $\mathbf{A}$  is square and all the  $\sigma_i$  are distinct, then the left and right singular vectors are uniquely determined up to complex signs (i.e., complex scalar factors of absolute value 1).*

## Theorem 4 (Real SVD)

Every matrix  $\mathbf{A} \in \mathbb{R}^{m \times n}$  has a real singular value decomposition.

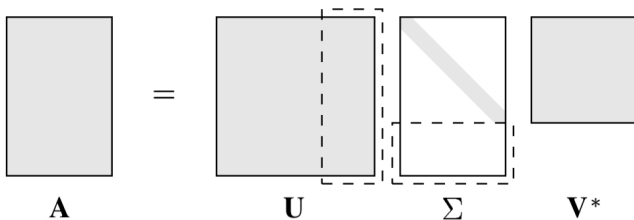
### 1.1. Variants of SVD

- Reduced SVD (the case  $m \geq n$ ):

$$\mathbf{A} = \mathbf{U}_n \mathbf{\Sigma}_n \mathbf{V}^*$$

where

$$\mathbf{U}_n = [\mathbf{u}_1 \quad \mathbf{u}_2 \quad \cdots \quad \mathbf{u}_n], \quad \mathbf{\Sigma}_n = \text{diag}\{\sigma_1, \sigma_2, \dots, \sigma_n\}.$$



- Rank SVD or compact SVD or condensed SVD:

$$\mathbf{A} = [\mathbf{U}_r \quad \mathbf{U}_c] \begin{bmatrix} \mathbf{\Sigma}_r & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{bmatrix} \begin{bmatrix} \mathbf{V}_r^* \\ \mathbf{V}_c^* \end{bmatrix} = \mathbf{U}_r \mathbf{\Sigma}_r \mathbf{V}_r^* = \sum_{i=1}^r \sigma_i \mathbf{u}_i \mathbf{v}_i^*,$$

where  $r = \text{rank}(\mathbf{A})$ ,

$$\mathbf{U}_r = [\mathbf{u}_1 \quad \mathbf{u}_2 \quad \cdots \quad \mathbf{u}_r], \quad \mathbf{U}_c = [\mathbf{u}_{r+1} \quad \mathbf{u}_{r+2} \quad \cdots \quad \mathbf{u}_m],$$

$$\mathbf{V}_r = [\mathbf{v}_1 \quad \mathbf{v}_2 \quad \cdots \quad \mathbf{v}_r], \quad \mathbf{V}_c = [\mathbf{v}_{r+1} \quad \mathbf{v}_{r+2} \quad \cdots \quad \mathbf{v}_n],$$

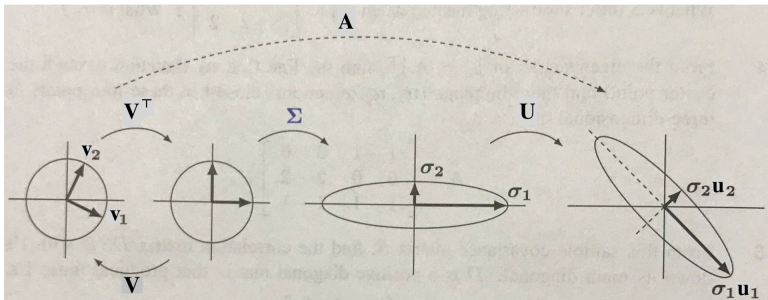
and

$$\mathbf{\Sigma}_r = \begin{bmatrix} \sigma_1 & & & \\ & \sigma_2 & & \\ & & \ddots & \\ & & & \sigma_r \end{bmatrix}.$$

## 1.2. Geometric observation

- The image of the unit sphere (in the 2-norm) of  $\mathbb{C}^n$  under any  $m \times n$  matrix is a hyperellipse of  $\mathbb{C}^m$ .

For example,  $2 \times 2$  real matrix  $\mathbf{A}$



**SVD of a matrix can not be emphasized too much!**

## 2. Matrix properties via SVD: $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^*$

- 2-norm

$$\|\mathbf{A}\|_2 = \sigma_1 = \|\mathbf{A}^*\|_2 = \|\mathbf{A}^\top\|_2 = \|\overline{\mathbf{A}}\|_2$$

- F-norm

$$\|\mathbf{A}\|_F = \sqrt{\sigma_1^2 + \sigma_2^2 + \cdots + \sigma_r^2} = \|\mathbf{A}^*\|_F = \|\mathbf{A}^\top\|_F = \|\overline{\mathbf{A}}\|_F$$

- *range* or *column space* of  $\mathbf{A} \in \mathbb{C}^{m \times n}$ , spanned by the columns of  $\mathbf{A}$

$$\begin{aligned}\text{range}(\mathbf{A}) &:= \{\mathbf{y} \in \mathbb{C}^m \mid \exists \mathbf{x} \in \mathbb{C}^n \text{ s.t. } \mathbf{y} = \mathbf{A}\mathbf{x}\} \\ &= \text{span}\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_r\} = \text{range}(\mathbf{U}_r)\end{aligned}$$

- *kernel* or *nullspace* of  $\mathbf{A} \in \mathbb{C}^{m \times n}$

$$\begin{aligned}\text{null}(\mathbf{A}) &:= \{\mathbf{x} \in \mathbb{C}^n \mid \mathbf{A}\mathbf{x} = \mathbf{0}\} \\ &= \text{span}\{\mathbf{v}_{r+1}, \mathbf{v}_{r+2}, \dots, \mathbf{v}_n\} = \text{range}(\mathbf{V}_c)\end{aligned}$$

- Range and nullspace of  $\mathbf{A}^*$ :

$$\begin{aligned}\text{range}(\mathbf{A}^*) &= \text{span}\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_r\} = \text{null}(\mathbf{A})^\perp \\ \text{null}(\mathbf{A}^*) &= \text{span}\{\mathbf{u}_{r+1}, \mathbf{u}_{r+2}, \dots, \mathbf{u}_m\} = \text{range}(\mathbf{A})^\perp\end{aligned}$$

- Relations between the four subspaces

$$\text{range}(\mathbf{A}^*) \perp \text{null}(\mathbf{A}), \quad \text{range}(\mathbf{A}^*) + \text{null}(\mathbf{A}) = \mathbb{C}^n$$

$$\text{range}(\mathbf{A}) \perp \text{null}(\mathbf{A}^*), \quad \text{range}(\mathbf{A}) + \text{null}(\mathbf{A}^*) = \mathbb{C}^m$$

- The absolute value of the determinant of  $\mathbf{A} \in \mathbb{C}^{m \times m}$ :

$$|\det(\mathbf{A})| = \prod_{i=1}^m \sigma_i$$

- If  $\mathbf{A}$  is Hermitian, i.e.,  $\mathbf{A} = \mathbf{A}^*$ , then

the singular values are the absolute values of the eigenvalues.

- An eigendecomposition via SVD:

$$\begin{bmatrix} \mathbf{0} & \mathbf{A} \\ \mathbf{A}^* & \mathbf{0} \end{bmatrix} = \mathbf{Q} \begin{bmatrix} \Sigma_r & & & \\ & -\Sigma_r & & \\ & & \mathbf{0} & \\ & & & \mathbf{0} \end{bmatrix} \mathbf{Q}^*, \mathbf{Q} = \begin{bmatrix} \frac{\mathbf{U}_r}{\sqrt{2}} & \frac{\mathbf{U}_r}{\sqrt{2}} & \mathbf{U}_c & \mathbf{0} \\ \frac{\mathbf{V}_r}{\sqrt{2}} & \frac{-\mathbf{V}_r}{\sqrt{2}} & \mathbf{0} & \mathbf{V}_c \end{bmatrix}.$$

- A random square matrix is “always” nonsingular. Or more general, a random rectangular matrix is “always” of full rank. Why?

## 2.1. Low-rank approximation (LRA)

### Theorem 5 (Eckart-Young-Mirski)

Assume that  $\mathbf{A} \in \mathbb{C}^{m \times n}$  has rank SVD  $\mathbf{A} = \mathbf{U}_r \mathbf{\Sigma}_r \mathbf{V}_r^*$ . For any integer  $k$  with  $1 \leq k < r = \text{rank}(\mathbf{A})$ , define  $\mathbf{A}_k = \sum_{i=1}^k \sigma_i \mathbf{u}_i \mathbf{v}_i^*$ . Then

$$\|\mathbf{A} - \mathbf{A}_k\|_2 = \sigma_{k+1} = \min_{\substack{\mathbf{B} \in \mathbb{C}^{m \times n}, \\ \text{rank}(\mathbf{B}) \leq k}} \|\mathbf{A} - \mathbf{B}\|_2,$$

and

$$\|\mathbf{A} - \mathbf{A}_k\|_F = \sqrt{\sigma_{k+1}^2 + \cdots + \sigma_r^2} = \min_{\substack{\mathbf{B} \in \mathbb{C}^{m \times n}, \\ \text{rank}(\mathbf{B}) \leq k}} \|\mathbf{A} - \mathbf{B}\|_F.$$

- **Discussion:** Is the minimizer in Theorem 5 unique?

## Proof of Theorem 5.

- Suppose there is some  $\mathbf{B} \in \mathbb{C}^{m \times n}$  with  $\text{rank}(\mathbf{B}) \leq k < r$  such that

$$\|\mathbf{A} - \mathbf{B}\|_2 < \sigma_{k+1}.$$

It follows that  $\dim(\text{null}(\mathbf{B})) = n - \text{rank}(\mathbf{B}) \geq n - k$ . Thus there exists an  $(n - k)$ -dimensional subspace  $\mathcal{W} \subseteq \text{null}(\mathbf{B})$ . For any nonzero  $\mathbf{x} \in \mathcal{W}$ , we have

$$\|\mathbf{A}\mathbf{x}\|_2 = \|(\mathbf{A} - \mathbf{B})\mathbf{x}\|_2 \leq \|\mathbf{A} - \mathbf{B}\|_2 \|\mathbf{x}\|_2 < \sigma_{k+1} \|\mathbf{x}\|_2.$$

Let  $\mathcal{V} = \text{span}\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_{k+1}\}$ . For any  $\mathbf{x} \in \mathcal{V}$ , we have

$$\|\mathbf{A}\mathbf{x}\|_2 = \|\mathbf{A}\mathbf{V}_{k+1}\mathbf{y}\|_2 = \|\mathbf{U}_{k+1}\mathbf{\Sigma}_{k+1}\mathbf{y}\|_2 = \|\mathbf{\Sigma}_{k+1}\mathbf{y}\|_2 \geq \sigma_{k+1} \|\mathbf{x}\|_2.$$

Since  $\dim \mathcal{W} + \dim \mathcal{V} = (n - k) + (k + 1) > n$ , there must be a nonzero vector lying in both, and this is a contradiction.

- Case  $\|\cdot\|_{\text{F}}$ : Generalized Inverses: Theory and Applications, 2nd edition, Adi Ben-Israel and Thomas N.E. Greville, Page 213.  $\square$

## Application of low-rank approximation: image compression

- An image can be represented as a matrix. For example, typical grayscale images consist of a rectangular array of pixels,  $m$  in the vertical direction,  $n$  in the horizontal direction. The color of each of those pixels is denoted by a single number, an integer between 0 (black) and 255 (white). (This gives  $2^8 = 256$  different shades of gray for each pixel. Color images are represented by three such matrices: one for red, one for green, and one for blue. Thus each pixel in a typical color image takes  $(2^8)^3 = 2^{24}$  shades.)
- The objective of image compression is to reduce irrelevance and redundancy of the image data in order to be able to store or transmit data in an efficient form.
- Low-rank SVD approximation is a good candidate. (Note: jpeg compression algorithm uses similar idea, on subimages)

Image Compression with Singular Value Decomposition Demo

### 3. Moore–Penrose pseudoinverse

- Let  $\mathbf{A} \in \mathbb{C}^{m \times n}$  have an SVD (rank form)  $\mathbf{A} = \mathbf{U}_r \mathbf{\Sigma}_r \mathbf{V}_r^*$ . The *Moore–Penrose pseudoinverse* of  $\mathbf{A}$ , denoted by  $\mathbf{A}^\dagger$ :

$$\mathbf{A}^\dagger := \mathbf{V}_r \mathbf{\Sigma}_r^{-1} \mathbf{U}_r^* = \sum_{i=1}^r \frac{1}{\sigma_i} \mathbf{v}_i \mathbf{u}_i^*.$$

- The matrix  $\mathbf{A}^\dagger$  is the *unique* matrix satisfying the four equations

$$\mathbf{A} \mathbf{X} \mathbf{A} = \mathbf{A}, \quad \mathbf{X} \mathbf{A} \mathbf{X} = \mathbf{X}, \quad (\mathbf{A} \mathbf{X})^* = \mathbf{A} \mathbf{X}, \quad (\mathbf{X} \mathbf{A})^* = \mathbf{X} \mathbf{A}.$$

For a proof, see Page 122 of [Numerical Linear Algebra \(in Chinese\)](#) by Zhihao Cao.

- If  $\mathbf{A}$  has full column rank, then

$$\mathbf{A}^\dagger = (\mathbf{A}^* \mathbf{A})^{-1} \mathbf{A}^*.$$

If  $\mathbf{A}$  has full row rank, then

$$\mathbf{A}^\dagger = \mathbf{A}^* (\mathbf{A} \mathbf{A}^*)^{-1}.$$

#### 4. A wonderful reference

- Zhihua Zhang (arXiv:1510.08532)

The singular value decomposition, applications and beyond

#### 5. Another proof of Theorem 5

- Holger Wendland

Numerical Linear Algebra An Introduction

Cambridge University Press, 2018.

See Page 295, Theorem 7.41.

#### 6. Computationally more feasible methods for LRA

- Adaptive cross approximation (ACA)

See Page 297 of Numerical Linear Algebra An Introduction.

- Joel A. Tropp and Robert J. Webber (arXiv:2306.12418)

Randomized algorithms for low-rank matrix approximation:  
Design, analysis, and applications